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B

magnitude of E at P due to $+64Q$

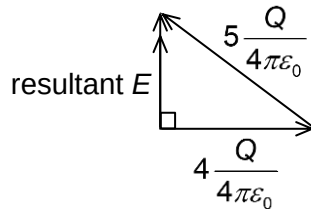
$$= \frac{64Q}{4\pi\epsilon_0 (4.0)^2} = 4 \frac{Q}{4\pi\epsilon_0}$$

magnitude of E at P due to $-125Q$

$$= \frac{125Q}{4\pi\epsilon_0 (5.0)^2} = 5 \frac{Q}{4\pi\epsilon_0}$$

These two E vectors form a right-angle triangle, with the resultant E pointing upwards

with magnitude $\sqrt{\left(5 \frac{Q}{4\pi\epsilon_0}\right)^2 - \left(4 \frac{Q}{4\pi\epsilon_0}\right)^2} = 3 \frac{Q}{4\pi\epsilon_0}$.



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A